

Technical Document #1013: Natural Language Processing - Comprehensive Overview

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The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Language models predict the probability of a sequence of words.
- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.
- Question answering systems extract answers from a given context.